



Online Shopping Analysis

Uncovering insights from **52,955 purchases** to guide strategic business decisions

Dataset at a Glance

52,955

Total Purchases

21

Columns

31

Missing Values

Across core columns

400

Coupon Nulls

Code & Discount %

Key Data Features

Customer Demographics

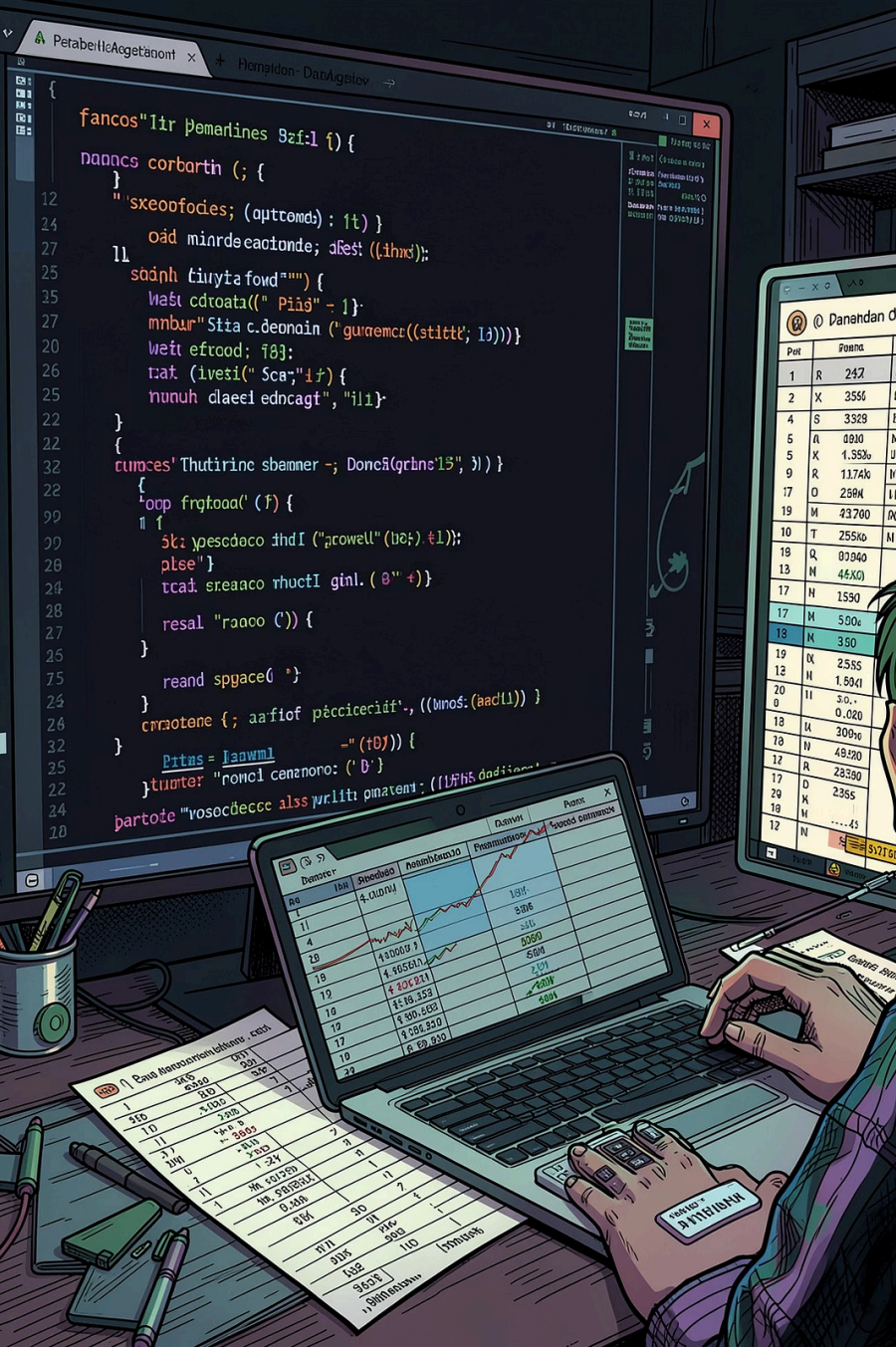
ID, Gender, Location, Tenure Months

Purchase Details

Transaction ID, SKU, Category, Price, GST, Delivery

Shopping Behavior

Coupon Status, Code, Discount %



Data Preparation in Python

01

Load & Explore

Imported via pandas; `df.info()` and `describe()`

02

Handle Missing Data

Removed 31 nulls; imputed "No coupon" for 400 records

03

Standardize Columns

Renamed to `snake_case` for readability

SQL Analysis: Financials & Sales

	Total_revenue	Total_discount_amount	Revenue_after_discount	Total_tax_amount	Revenue_after_taxes	Total_delivery_charges	Net_profit
▶	4670794.62	925958.24	3744836.38	514539.12	3230297.26	556635.07	2673662.19

Financial Position

Website financial statement for the last month

Total Sales Volume

Overall number of transactions completed

Revenue by Category

Top 5 highest-revenue product categories

SQL Analysis: Products & Customers

	gender	Number_of_customers
▶	F	934
	M	534

Top Products by Price

5 most expensive items sold

Total Customers

Unique buyers last month

Gender Split

Male vs Female purchasers

Loyalty Program

Customers with 12+ months tenure

SQL Analysis: Geography & Segments

	customer_segment	Number_of_customers
▶	Loyal	1169
	Existing	299

Customers by State

Purchase count per location

Revenue by State

Total revenue per state

Customer Segments

New · Existing · Loyal — based on tenure

SQL Analysis: Discounts & Categories

	coupon_code	total_orders
▶	SALE20	6373
	SALE30	5915
	SALE10	5838
	ELEC10	4826
	ELEC30	4647

Top Coupon Codes

5 most-used promo codes by customers

Sales by Category

Total sales volume per product category

Online Shopping Dashboard

Product Category Slicer

All

State Location Slicer

All

Number of customers

1.47K

Number of product sold

238.03K

Total revenue

4.67M

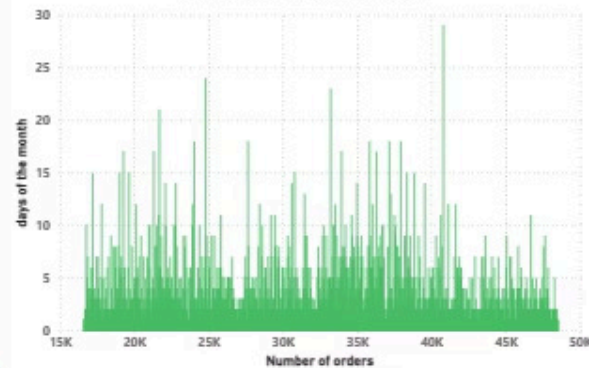
Total orders

53K

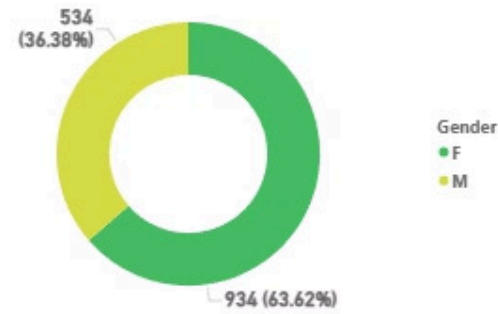
Top state of revenue

Chicago

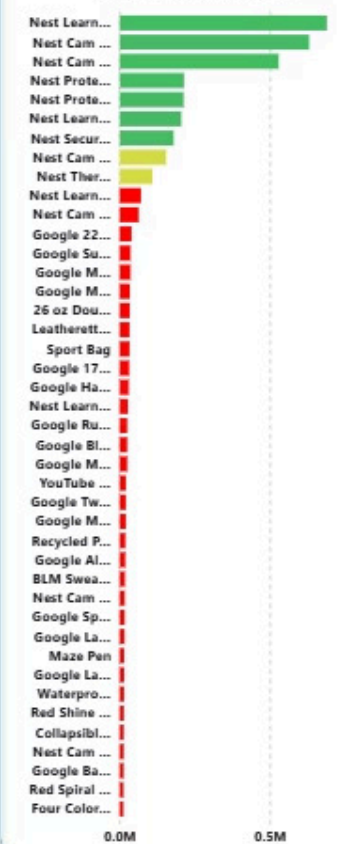
Orders over the month



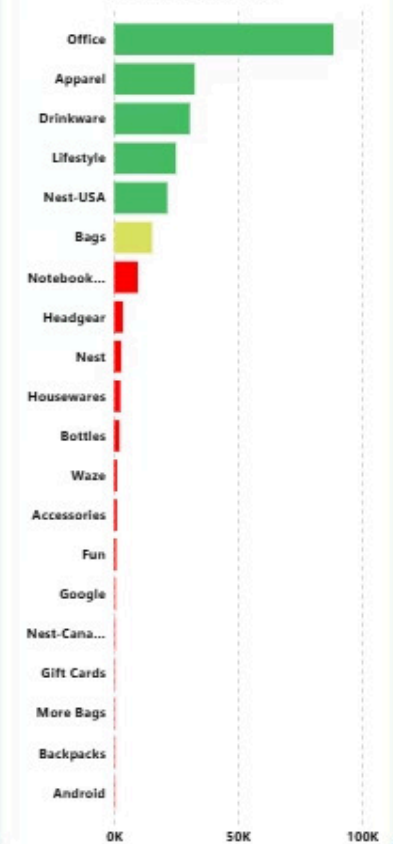
% of Male vs Female customers



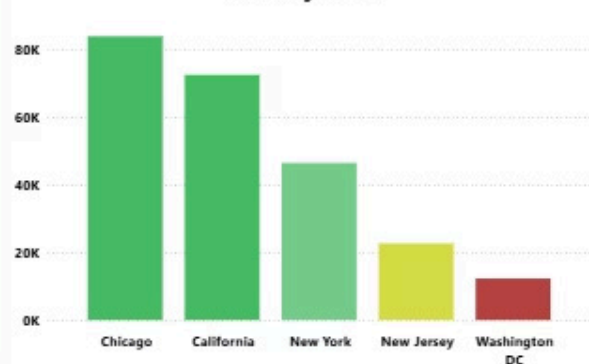
Revenue by product



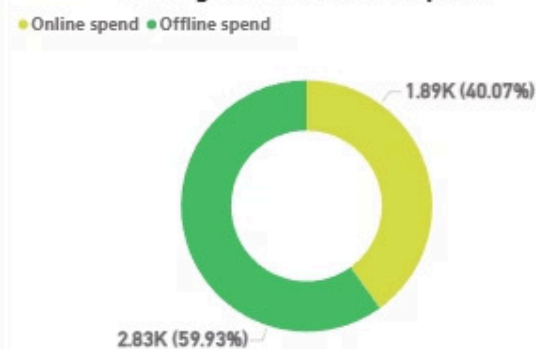
Sales by Category



Sales by State



Average online vs offline spend



Interactive Power BI Dashboard

All insights visualized in a single interactive dashboard for stakeholders

Business Recommendations

Premium Offers

Promote exclusive benefits for low-revenue products

Loyalty Programs

Reward tenure buyers to advance to "Loyal" segment

Review Discounts

Balance sales boosts with margin control

Product Positioning

Highlight top-rated and best-selling products

Targeted Marketing

Focus on high-revenue products & top promo codes; expand retailer & influencer partnerships